

The Ephemeris

September 2021

Volume 32 Number 03- The Official Publication of the San Jose Astronomical Association



September 2021 - December 2021 Events

Schedule subject to change due to Covid-19 restrictions.

Please check our website for up to date information.

Board & General Meetings

Saturday 9/18, 10/23, 11/7, 12/5

Board Meetings: 6 -7:30pm

General Meetings: 8:00-9:30pm

Fix-It Day (2-4pm)

Currently on hold due to COVID-19

Solar Sunday (2-4pm)

Sunday 9/12, 10/3, 11/8, 12/6

Intro to the Night Sky Class & Hough Park 1Q In-Town Star Party

Currently on hold due to COVID-19

RCDO Starry Nights Star Party

Currently on hold due to COVID-19

Imaging SIG Mtg

Tuesday 9/21, 10/19, 11/17, 12/15

Please refer to the SJAA Website home page

(www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Photo by SJAA Member: Nikola Nikolov
Jan. 26, 2021, Feb. 7, 2021
Location: Chile

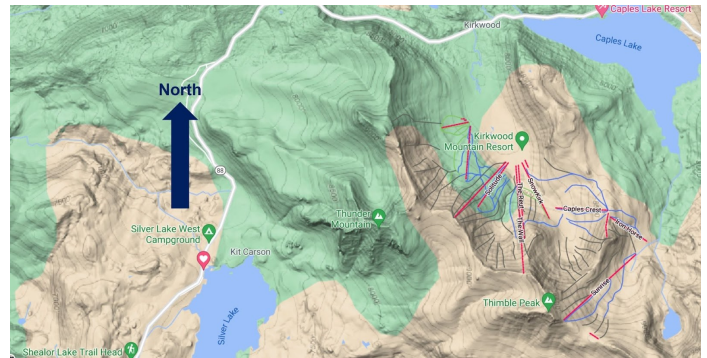
An exploratory trip to Carson Pass - Observation Report

Credit for text and images - Nikola Nikolov

Motivated by the increasing light pollution in the Bay Area, I wanted to find a new location for amateur astronomy in Sierra's, which is within a 3 hours drive from San Jose and has an exceptionally dark sky. Currently, Pinnacles National Park in San Benito County offers the best possible conditions with sky-glow brightness Bortle 3 and about a two hours drive from San Jose. South of Pinnacles National Park is Lake San Antonio near Paso Robles, which is about 3 hours drive and Bortle 2 sky-glow brightness. Lake San Antonio is famous for the CalStar astronomy party, which is usually organized in April and October each year. The question is, is there a 3rd location similar to Pinnacles national park and Lake San Antonio, where both observer and imaging astronomers can set up telescope equipment, and it is still within 3 hours drive from Bay Area. Some of the best possibilities for such locations are in the Sierra Mountains, but not many people know them. What I particularly like in the Sierra's is the high elevation terrain, which is a major advantage vs Pinnacles and Lake San Antonio. As we know, most of the professional research observatories in the world are located on very high mountain elevations with cold and dry climates.

On July 3rd I made a small family trip to Carson Pass Region in Eldorado county, California. During the trip I was planning to explore new locations in Sierra Nevada mountains which are good for telescope setup, both observing and astrophotography. Yet I didn't carry any telescopes with me, because I didn't know if the locations were suitable for them. I had only a DSLR Canon camera with 50mm lens, and shutter to capture the Milky Way. The locations I explored were along Highway 88, known as Carson Pass. The road climbs close to 8000ft in the mountains and reaches Kirkwood ski resort. During the winter driving along highway 88 is not recommended for vehicles other than AWD or 4WD, as well as good tires and snow chains (just in case). Summer time the road is accessible to all vehicles. Since elevation of terrain is near or above 8000ft, it is an excellent location for meteor shower observing. The locations for astronomy I was exploring were near Silver Lake and Caples Lake. They are approximately within a 3 hours drive from the Bay Area, which is similar to driving to Lake San Antonio near Paso Robles. Silver and Caples Lakes are about 40 minutes drive from South Lake Tahoe by taking Highway 89 South, and then turn right on Highway 88 towards Kirkwood Ski resort.

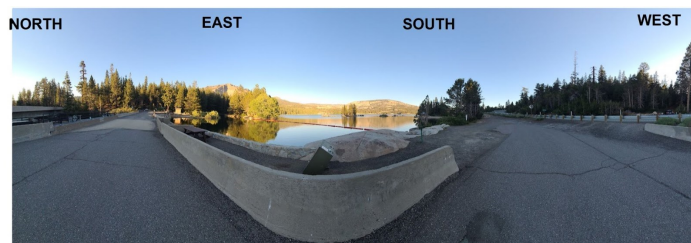
Both Silver Lake and Caples Lake offer wide spaces suitable for car parking and telescope setup, but Silver Lake seems to be the better of both places. The spot I liked for the astronomy-rig set up near Silver Lake is the one next to Kit Carson Lodge. The site is a water dam for the Silver Lake with a parking lot for daytime fishing and free Wi-Fi provided by Kit Carson Lodge. The water dam orientation is North-East/South West.



Silver and Caples Lakes with Kirkwood Ski resort in the middle. The peak in the center of the map is called "Thunder Mountain". The peak creates beautiful mirror effects in Silver Lake during summer as shown in the photography below.



The location near Kit Carson Lodge I chose for nightscape photography. The arrow denotes the Silver Lake water dam which can accommodate cars and telescope rigs



Silver Lake water-dam and the approximate orientations

While I was there with we met a member from the Sacramento Valley Astronomy Society (SVAS). She revealed that Silver Lake water-dam has been used by SVAS for organized astronomy events before the COVID19 pandemic.

Therefore, I would recommend a few SJAA enthusiasts to make a future trip to Silver Lake with telescope rigs. Since we are in late August and Highway 88 becomes snowy in winter, one can plan such trip either before the end of September 2021 or May-June 2022.

Fizzing Sodium Could Explain Asteroid Phaethon's Cometlike Activity

Credit: NASA



This illustration depicts asteroid Phaethon being heated by the Sun. The asteroid's surface gets so hot that sodium inside Phaethon's rock may vaporize and vent into space, causing it to brighten like a comet and dislodge small pieces of rocky debris.

Credits: NASA/JPL-Caltech/IPAC

Models and lab tests suggest the asteroid could be venting sodium vapor as it orbits close to the Sun, explaining its increase in brightness.

As a comet zooms through the inner solar system, the Sun heats it, causing ices below the surface to vaporize into space. The venting vapor dislodges dust and rock, and the gas creates a bright tail that can extend millions of miles from the nucleus like an ethereal veil.

Whereas comets contain lots of different ices, asteroids are mainly rock and not known for producing such majestic displays. But a new study examines how near-Earth asteroid Phaethon may in fact exhibit cometlike activity, despite lacking significant quantities of ice.

Known to be the source of the annual Geminid meteor shower, the 3.6-mile-wide (5.8 kilometer-wide) asteroid brightens as it gets close to the Sun. Comets typically behave like this: When they heat up, their icy surfaces vaporize, causing them to become more active and brighten as the venting gases and dust scatter more sunlight. But what is causing Phaethon to brighten if not vaporizing ices?

The culprit could be sodium. As the new study's authors explain, Phaethon's elongated, 524-day orbit takes the object well within the orbit of Mercury, during which time the Sun heats the asteroid's surface up to about 1,390 degrees Fahrenheit (750 degrees Celsius). With such a warm orbit, any water, carbon dioxide, or carbon monoxide ice near the asteroid's surface would have been baked off long ago. But at that temperature, sodium may be fizzing from the asteroid's rock and into space.

"Phaethon is a curious object that gets active as it approaches the Sun," said study lead Joseph Masiero, a scientist at IPAC, a research organization at Caltech. "We know it's an asteroid and the source of the Geminids. But it contains little to no ice, so we were intrigued by the possibility that sodium, which is relatively plentiful in asteroids, could be the element driving this activity."

Asteroid-Meteor Connection

Masiero and his team were inspired by observations of the Geminids. When meteoroids – small pieces of rocky debris from space – streak through Earth's atmosphere as meteors, they disintegrate. But before they do, friction with the atmosphere causes the air surrounding the meteoroids to reach thousands of degrees, generating light. The color of this light represents the elements they contain. Sodium, for example, creates an orange tinge. The Geminids are known to be low in sodium.

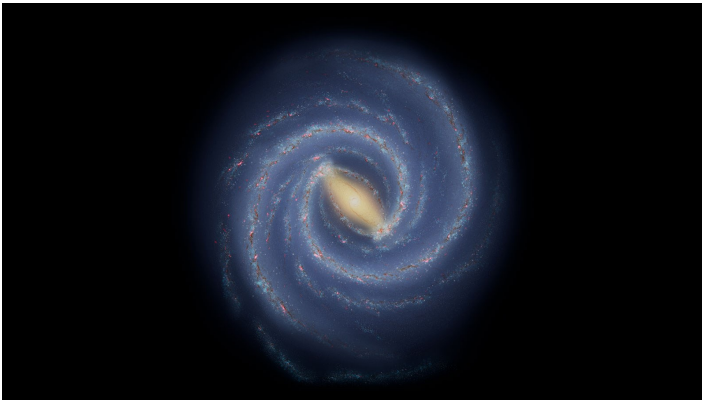
Until now, it was assumed that these small pieces of rock somehow lost their sodium after leaving the asteroid. This new study suggests that the sodium may actually play a key role in ejecting the Geminid meteoroids from Phaethon's surface.

The researchers think that as the asteroid approaches the Sun, its sodium heats up and vaporizes. This process would have depleted the surface of sodium long ago, but sodium within the asteroid still heats up, vaporizes, and fizzes into space through cracks and fissures in Phaethon's outermost crust. These jets would provide enough oomph to eject the rocky debris off its surface. So the fizzing sodium could explain not only the asteroid's cometlike brightening, but also how the Geminid meteoroids would be ejected from the asteroid and why they contain little sodium.

"Asteroids like Phaethon have very weak gravity, so it doesn't take a lot of force to kick debris from the surface or dislodge rock from a fracture," said Björn Davidsson, a scientist at NASA's Jet Propulsion Laboratory in Southern California and a co-author of the study. "Our models suggest that very small quantities of sodium are all that's needed to do this – nothing explosive, like the erupting vapor from an icy comet's surface; it's more of a steady fizz."

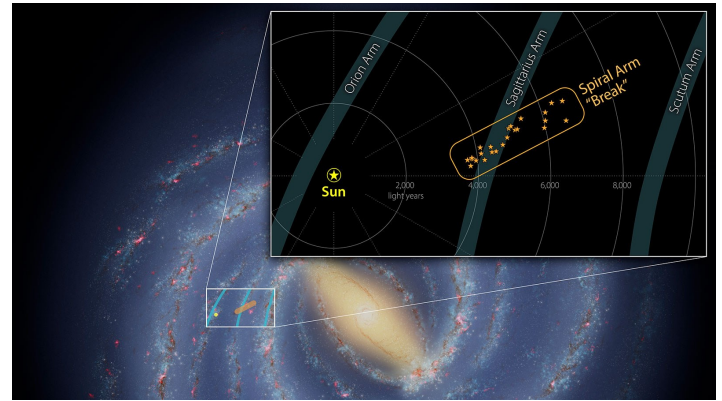
Astronomers Find a ‘Break’ in One of the Milky Way’s Spiral Arms

Credit: NASA



This illustration shows astronomers’ current understanding of the large-scale structure of the Milky Way. Stars and star-forming regions are largely grouped into spiral arms. Measuring the shape, size, and number of spiral arms is a challenge because Earth is located inside the galaxy.

Credits: NASA/JPL-Caltech



A contingent of stars and star-forming clouds was found jutting out from the Milky Way's Sagittarius Arm. The inset shows the size of the structure and distance from the Sun. Each orange star shape indicates star-forming regions that may contain anywhere from dozens to thousands of stars

Credits: NASA/JPL-Caltech

The newly discovered feature offers insight into the large-scale structure of our galaxy, which is difficult to study from Earth’s position inside it.

Scientists have spotted a previously unrecognized feature of our Milky Way galaxy: A contingent of young stars and star-forming gas clouds is sticking out of one of the Milky Way’s spiral arms like a splinter poking out from a plank of wood. Stretching some 3,000 light-years, this is the first major structure identified with an orientation so dramatically different than the arm’s.

Astronomers have a rough idea of the size and shape of the Milky Way’s arms, but much remains unknown: They can’t see the full structure of our home galaxy because Earth is inside it. It’s akin to standing in the middle of Times Square and trying to draw a map of the island of Manhattan. Could you measure distances precisely enough to know if two buildings were on the same block or a few streets apart? And how could you hope to see all the way to the tip of the island with so many things in your way?

To learn more, the authors of the new study focused on a nearby portion of one of the galaxy’s arms, called the Sagittarius Arm. Using NASA’s Spitzer Space Telescope prior to its retirement in January 2020, they sought out newborn stars, nestled in the gas and dust clouds (called nebulae) where they form. Spitzer detects infrared light that can penetrate those clouds, while visible light (the kind human eyes can see) is blocked.

Young stars and nebulae are thought to align closely with the shape of the arms they reside in. To get a 3D view of the arm segment, the scientists used the latest data release from the ESA (European Space Agency) Gaia mission to measure the precise distances to the stars. The combined data revealed that the long, thin structure associated with the Sagittarius Arm is made of young stars moving at nearly the same velocity and in the same direction through space.

“A key property of spiral arms is how tightly they wind around a galaxy,” said Michael Kuhn, an astrophysicist at Caltech and lead author of the new paper. This characteristic is measured by the arm’s pitch angle. A circle has a pitch angle of 0 degrees, and as the spiral becomes more open, the pitch angle increases. “Most models of the Milky Way suggest that the Sagittarius Arm forms a spiral that has a pitch angle of about 12 degrees, but the structure we examined really stands out at an angle of nearly 60 degrees.”

Similar structures – sometimes called spurs or feathers – are commonly found jutting off the arms of other spiral galaxies. For decades scientists have wondered whether our Milky Way’s spiral arms are also dotted with these structures or if they are relatively smooth.

Continued on Page 5



Measuring the Milky Way

The newly discovered feature contains four nebulae known for their breathtaking beauty: the Eagle Nebula (which contains the Pillars of Creation), the Omega Nebula, the Trifid Nebula, and the Lagoon Nebula. In the 1950s, a team of astronomers made rough distance measurements to some of the stars in these nebulae and were able to infer the existence of the Sagittarius Arm. Their work provided some of the first evidence of our galaxy's spiral structure.

"Distances are among the most difficult things to measure in astronomy," said co-author Alberto Krone-Martins, an astrophysicist and lecturer in informatics at the University of California, Irvine and a member of the Gaia Data Processing and Analysis Consortium (DPAC). "It is only the recent, direct distance measurements from Gaia that make the geometry of this new structure so apparent."

In the new study, researchers also relied on a catalog of more than a hundred thousand newborn stars discovered by Spitzer in a survey of the galaxy called the Galactic Legacy Infrared Mid-Plane Survey Extraordinaire (GLIMPSE).

"When we put the Gaia and Spitzer data together and finally see this detailed, three-dimensional map, we can see that there's quite a bit of complexity in this region that just hasn't been apparent before," said Kuhn.



Shown here (from left) are the Eagle, Omega, Trifid, and Lagoon Nebulae, imaged by NASA's infrared Spitzer Space Telescope. These nebulae are part of a structure within the Milky Way's Sagittarius Arm that is poking out from the arm at a dramatic angle.

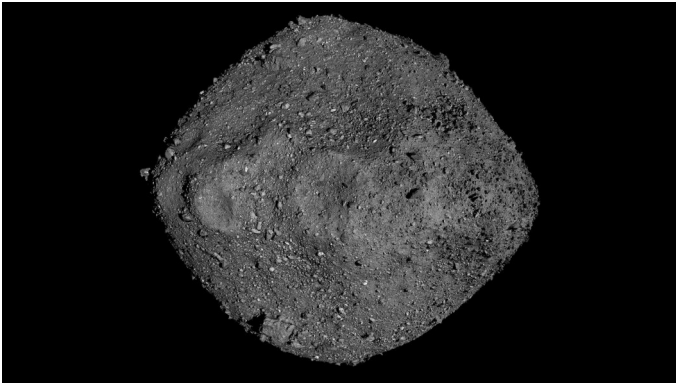
Credits: NASA/JPL-Caltech

Astronomers don't yet fully understand what causes spiral arms to form in galaxies like ours. Even though we can't see the Milky Way's full structure, the ability to measure the motion of individual stars is useful for understanding this phenomenon: The stars in the newly discovered structure likely formed around the same time, in the same general area, and were uniquely influenced by the forces acting within the galaxy, including gravity and shear due to the galaxy's rotation.

"Ultimately, this is a reminder that there are many uncertainties about the large-scale structure of the Milky Way, and we need to look at the details if we want to understand that bigger picture," said one of the paper's co-authors, Robert Benjamin, an astrophysicist at the University of Wisconsin-Whitewater and a principal investigator on the GLIMPSE survey. "This structure is a small piece of the Milky Way, but it could tell us something significant about the Galaxy as a whole."

NASA Spacecraft Provides Insight into Asteroid Bennu's Future Orbit

Credit: NASA



This mosaic of Bennu was created using observations made by NASA's OSIRIS-REx spacecraft that was in close proximity to the asteroid for over two years.

Credits: NASA/Goddard/University of Arizona

In a study released Wednesday, NASA researchers used precision-tracking data from the agency's Origins, Spectral Interpretation, Resource Identification, Security-Regolith Explorer (OSIRIS-REx) spacecraft to better understand movements of the potentially hazardous asteroid Bennu through the year 2300, significantly reducing uncertainties related to its future orbit, and improving scientists' ability to determine the total impact probability and predict orbits of other asteroids.

The study, titled "Ephemeris and hazard assessment for near-Earth asteroid (101955) Bennu based on OSIRIS-REx data," was published in the journal *Icarus*.

"NASA's Planetary Defense mission is to find and monitor asteroids and comets that can come near Earth and may pose a hazard to our planet," said Kelly Fast, program manager for the Near-Earth Object Observations Program at NASA Headquarters in Washington. "We carry out this endeavor through continuing astronomical surveys that collect data to discover previously unknown objects and refine our orbital models for them. The OSIRIS-REx mission has provided an extraordinary opportunity to refine and test these models, helping us better predict where Bennu will be when it makes its close approach to Earth more than a century from now."

In 2135, asteroid Bennu will make a close approach with Earth. Although the near-Earth object will not pose a danger to our planet at that time, scientists must understand Bennu's exact trajectory during that encounter in order to predict how Earth's gravity will alter the asteroid's path around the Sun – and affect the hazard of Earth impact.

Using NASA's Deep Space Network and state-of-the-art computer models, scientists were able to significantly shrink uncertainties in Bennu's orbit, determining its total impact probability through the year 2300 is about 1 in 1,750 (or 0.057%). The researchers were also able to identify Sept. 24, 2182, as the most significant single date in terms of a potential impact, with an impact probability of 1 in 2,700 (or about 0.037%).

Although the chances of it hitting Earth are very low, Bennu remains one of the two most hazardous known asteroids in our solar system, along with another asteroid called 1950 DA.

Before leaving Bennu May 10, 2021, OSIRIS-REx spent more than two years in close proximity to the asteroid, gathering information about its size (it is about one-third of a mile, or 500 meters, wide), shape, mass, and composition, while monitoring its spin and orbital trajectory. The spacecraft also scooped up a sample of rock and dust from the asteroid's surface, which it will deliver to Earth on Sept. 24, 2023, for further scientific investigation.

"The OSIRIS-REx data give us so much more precise information, we can test the limits of our models and calculate the future trajectory of Bennu to a very high degree of certainty through 2135," said study lead Davide Farnocchia, of the Center for Near Earth Object Studies (CNEOS), which is managed by NASA's Jet Propulsion Laboratory in Southern California. "We've never modeled an asteroid's trajectory to this precision before."

Gravitational keyholes

The precision measurements on Bennu help to better determine how the asteroid's orbit will evolve over time and whether it will pass through a "gravitational keyhole" during its 2135 close approach. These keyholes are areas in space that would set Bennu on a path toward a future impact with Earth if the asteroid were to pass through them at certain times, due to the effect of Earth's gravitational pull.

To calculate exactly where the asteroid will be during its 2135 close approach – and whether it might pass through a gravitational keyhole – Farnocchia and his team evaluated various types of small forces that may affect the asteroid as it orbits the Sun. Even the smallest force can significantly deflect its orbital path over time, causing it to pass through or completely miss a keyhole.

SJAA Member Astrophoto Gallery



Date: August 8 and 10 2021

Location: Mountain View, CA and Pinnacles East, Paicines, CA

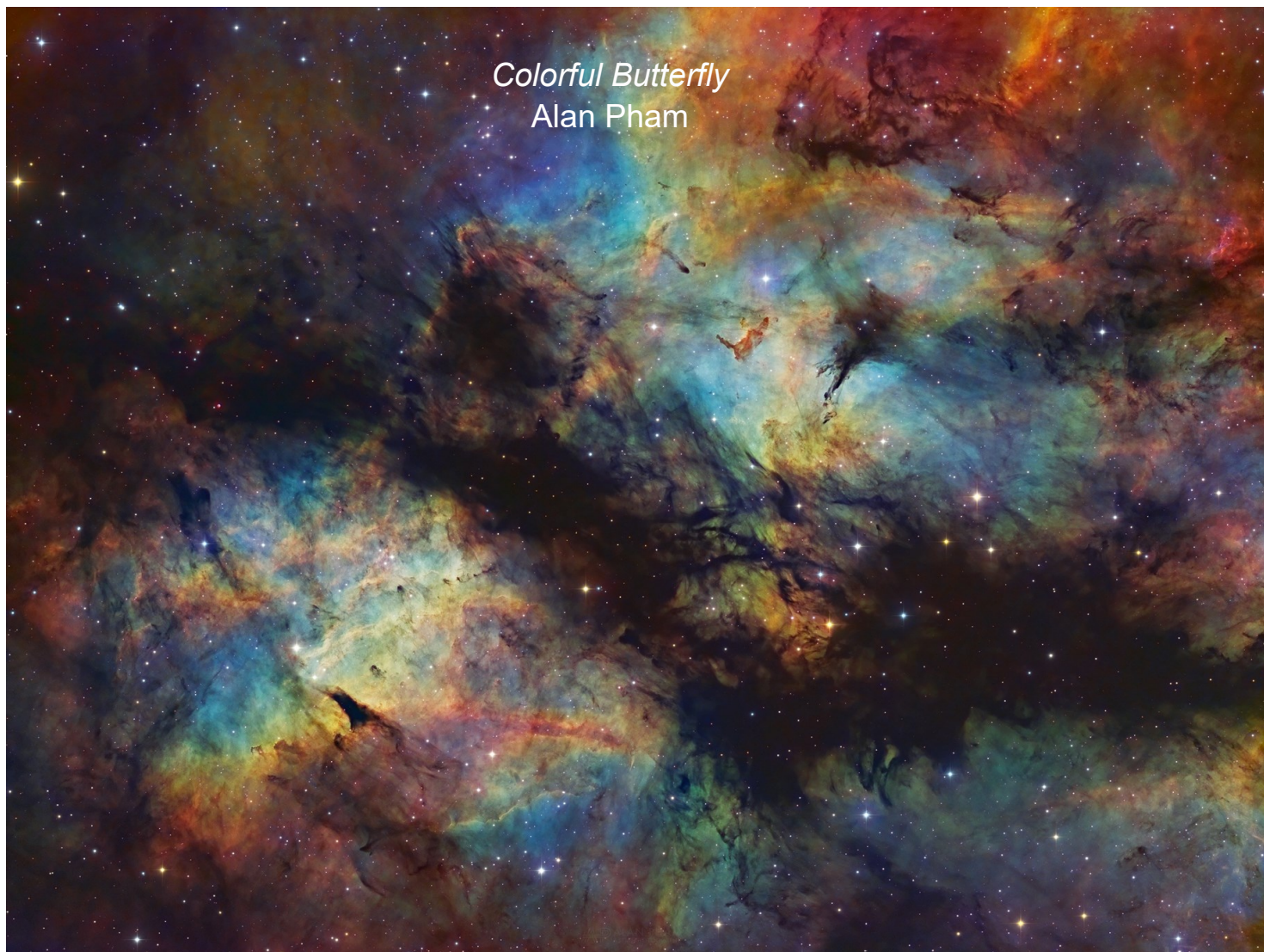
Imaging Telescope: Astro-Tech AT130EDT

Imaging Camera: ZWO ASI294MM-Pro

Mount: iOptron iEQ45 Pro

Software: N.I.N.A. Nighttime Imaging 'N' Astronomy, Pleiades Astrophoto PixInsight 1.8 Ripely

SJAA Member Astrophoto Gallery



Dates: June 26, 2020

Location: San Jose, CA

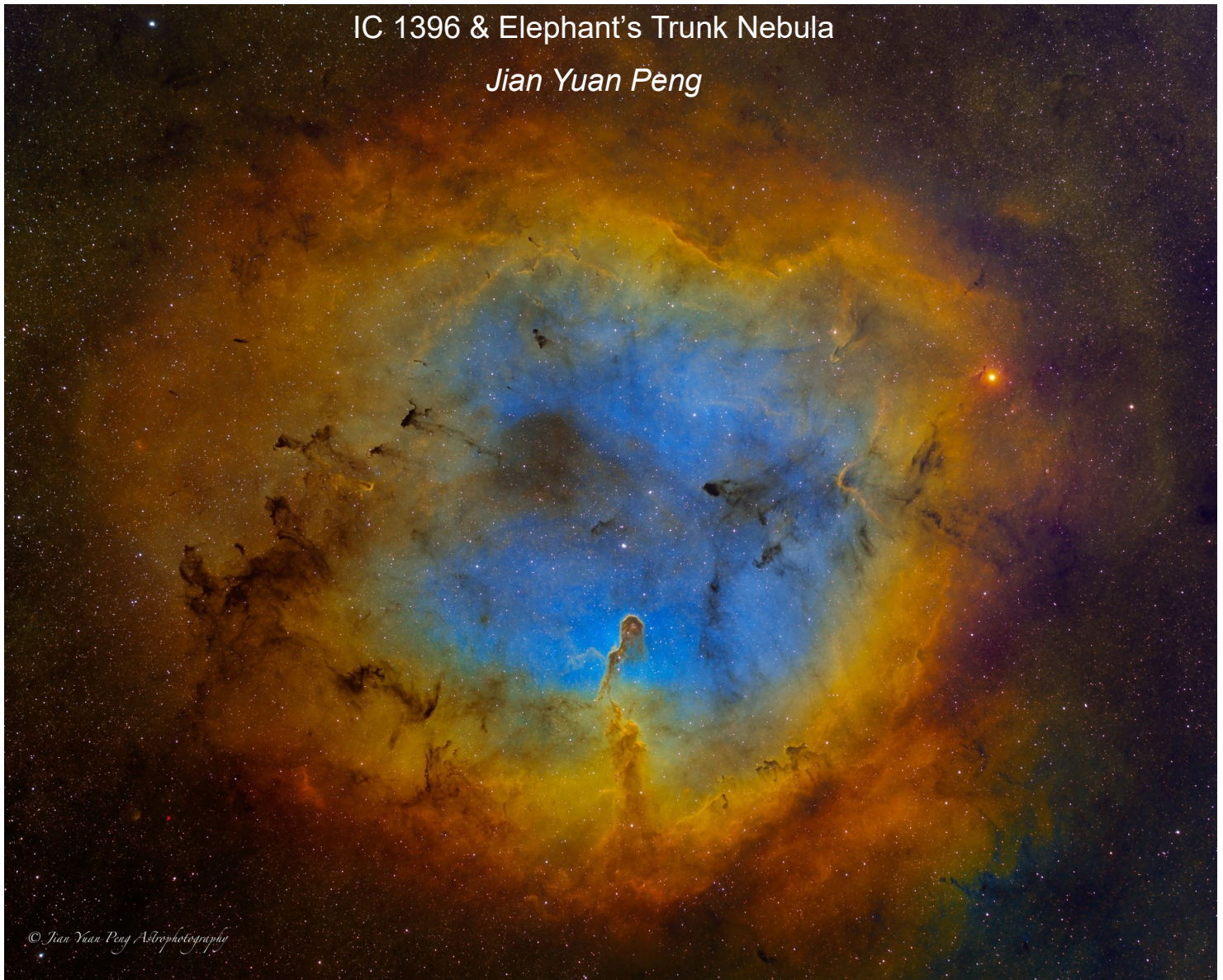
Telescope: Stellarvue SV4

Camera: ZWO ASI1600MM-Cool

Mount: Astro-Physics Mach1 GTO

Software: PhotoShop, PHD2, Sequence Generator Pro, PixInsight

SJAA Member Astrophoto Gallery



Date: August 13, 2021

Location: Palo Alto, CA

Telescope: Takahashi FSQ-106EDX4

Camera: ZWO ASI6200MM

Mount: iOptron CEM40EC

Software: PHD2, PixInsight, Lightroom 6.4, Sequence Generator Pro, Photoshop CS6, StarNet++

SJAA Member Astrophoto Gallery



Dates: July 23, 2021

Location: Menlo Park

Telescope: William Optics ZenithStar 61-ii

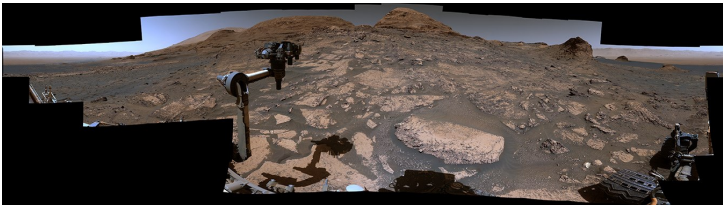
Camera: QHY5L-II-M

Mount: iOptron CEM40

Software: ASTAP, Lightroom, DeNoise AI, PixInsight, PHD2, SharpCap, APT, Stellarium

NASA's Curiosity Mars Rover Explores a Changing Landscape

Credit: NASA



NASA's Curiosity Mars rover used its Mast Camera, or Mastcam, to capture this 360-degree view on July 3, 2021, the 3,167th Martian day, or sol, of the mission. The panorama is made up of 129 individual images that were sent to Earth, after which they are stitched together.

Credits: NASA/JPL-Caltech/MSSS

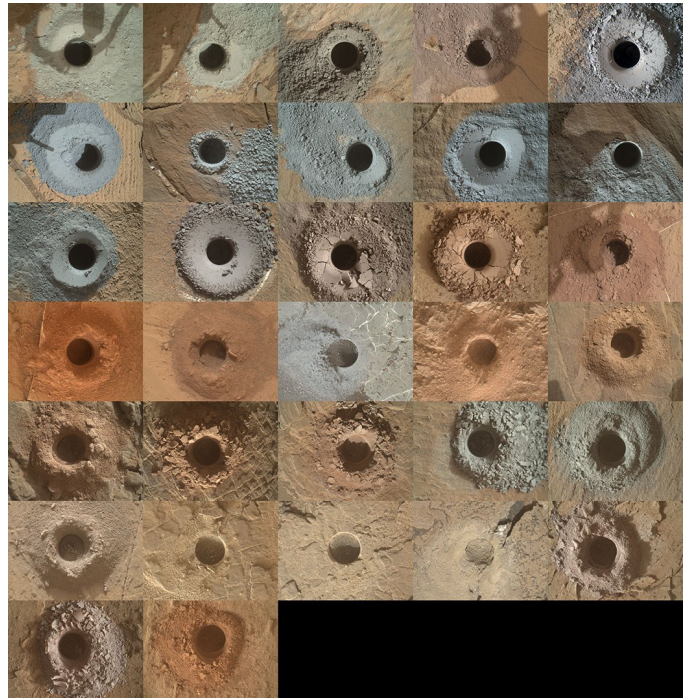
Images of knobby rocks and rounded hills are delighting scientists as NASA's Curiosity rover climbs Mount Sharp, a 5-mile-tall (8-kilometer-tall) mountain within the 96-mile-wide (154-kilometer-wide) basin of Mars' Gale Crater. The rover's Mast Camera, or Mastcam, highlights those features in a panorama captured on July 3, 2021 (the 3,167th Martian day, or sol, of the mission).

This location is particularly exciting: Spacecraft orbiting Mars show that Curiosity is now somewhere between a region enriched with clay minerals and one dominated by salty minerals called sulfates. The mountain's layers in this area may reveal how the ancient environment within Gale Crater dried up over time. Similar changes are seen across the planet, and studying this region up close has been a major long-term goal for the mission.

"The rocks here will begin to tell us how this once-wet planet changed into the dry Mars of today, and how long habitable environments persisted even after that happened," said Abigail Fraeman, Curiosity's deputy project scientist, at NASA's Jet Propulsion Laboratory in Southern California.

Nine Years on Mars

Curiosity landed nine years ago, on Aug. 5, 2012 PDT (Aug. 6, 2012 EDT), to study whether different Martian environments could have supported microbial life in the planet's ancient past, when lakes and groundwater existed within Gale Crater. The rover pulverizes rock samples with a drill on its robotic arm, then sprinkles the powder into the rover's chassis, where a pair of instruments determines which chemicals and minerals are present. Curiosity recently drilled its 32nd rock sample from a target nicknamed "Pontours" that will help detail the transition from the region of clay minerals to the one dominated by sulfates.



NASA's Curiosity Mars rover has used the drill on its robotic arm to take 32 rock samples to date. The Mars Hand Lens Imager (MAHLI), a camera on the end of the robotic arm, provided the images in this mosaic.

Credits: NASA/JPL-Caltech/MSSS

Because it's winter at Curiosity's location, the skies in the new panorama are relatively dust-free, providing a clear view all the way down to Gale Crater's floor. It's provided an opportunity for the mission team to reflect on the 16 miles (26 kilometers) Curiosity has driven during the mission.

"Landing day is still one of the happiest days of my professional career," said the mission's new project manager, Megan Richardson Lin of NASA's Jet Propulsion Laboratory in Southern California. Lin started working on Curiosity just before it launched, joining the surface operations team shortly afterward. She's held several roles on the mission since then. "We're driving a robot as it explores another planet. Seeing how new discoveries and scientific results guide each day's activities is extremely rewarding."



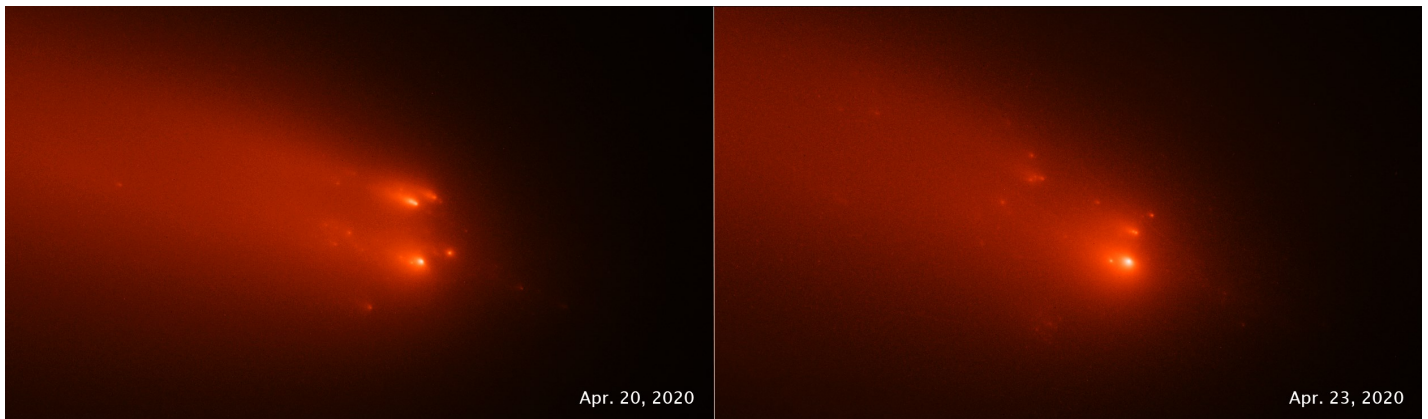
Kid Spot Jokes:

- ♦ The puzzled astronomy student spent all night wondering where the Sun went..
... but then it dawned on him
- ♦ How do astronomers organize a party?
They planet.

Fun Facts

Comet Atlas May Have Been a Blast From the Past

Credit: NASA



This pair of Hubble Space Telescope images of comet C/2019 Y4 (ATLAS), taken on April 20 and April 23, 2020, reveal the breakup of the solid nucleus of the comet. Hubble photos identify as many as 30 separate fragments. The comet was approximately 91 million miles from Earth when the images were taken. The comet may be a broken off piece of a larger comet that swung by the Sun 5,000 years ago. The comet has been artificially colored in this view to enhance details for analysis.

Credits: Science: NASA, ESA, Quanzhi Ye (UMD); Image Processing: Alyssa Pagan (STScI)

It's suspected that about 5,000 years ago a comet swept within 23 million miles of the Sun, closer than the innermost planet Mercury. The comet might have been a spectacular sight to civilizations across Eurasia and North Africa at the end of the Stone Age.

However, this nameless space visitor is not recorded in any known historical account. So how do astronomers know that there was such an interplanetary intruder?

Enter comet ATLAS (C/2019 Y4), which first appeared near the beginning of 2020.

Comet ATLAS, first detected by the Asteroid Terrestrial -impact Last Alert System (ATLAS), operated by the University of Hawaii, quickly met an untimely death in mid-2020 when it disintegrated into a cascade of small icy pieces.

In a new study using observations from NASA's Hubble Space Telescope, astronomer Quanzhi Ye of the University of Maryland in College Park, reports that ATLAS is a broken-off piece of that ancient visitor from 5,000 years ago. Why? Because ATLAS follows the same orbital "railroad track" as that of a comet seen in 1844. This means the two comets are probably siblings from a parent comet that broke apart many centuries earlier. The link between the two comets was first noted by amateur astronomer Maik Meyer.

Such comet families are common. The most dramatic visual example was in 1994 when the doomed comet Shoemaker-Levy 9 (SL9) was pulled into a string of pieces by Jupiter's gravitational pull. This "comet train" was short-lived. It fell piece by piece into Jupiter in July 1994.

From the Board of Directors

Announcements

- ◇ San Jose Astronomical Association is closely monitoring the coronavirus/COVID-19 situation in Santa Clara County. In accordance with guidance from Santa Clara County Health Department, SJAA leaders have been canceling many events, especially those that involve close contact and shared surfaces on telescopes. Some events, however, have successfully been moved to an online format. Please check our Meetup to determine if an event has been canceled or moved online. The board of directors will periodically review the status of the outbreak and the guidance from the county to determine the impact upon upcoming events.
- ◇ Peter Melhus is the new Secretary until the March 2022 board election.
- ◇ The Vice-President position is still open.
- ◇ SJAA's School Star Party program is planning to open its programs this fall. The program is seeking volunteers to help with scheduling schools, managing the calendar and recruiting volunteer astronomers. In addition you'll want to sign up soon to be a volunteer astronomer because It is so fun and fulfilling! For more information email Cat at ssp@sjaa.net.
- ◇ We are transitioning leadership of the School Star Party program. If you are interested in helping out, please email Cat at ssp@sjaa.net. It's loads of fun! With the continuing pandemic we are looking to include a virtual element to the parties. Cat is happy to share more.

Board Meeting Excerpts

June 26, 2021

In attendance

Swami, Sandy, Emi, Glenn, Ken, Wolf, Sukhada, Vini
Absent: Peter, Rob
Guests: Hy, Sheena, Carl, Mark, Erika, Kanch, Rishabh

Proposal for Houge Park

SJAA submitted a new proposal with the City's Parks Department for continued use of the Hall at Houge Park. The proposal has been accepted for the July 2021 – June 2022 term except for insurance.

Loaner Program Resurrection

SJAA is waiting to get access to Houge Park to start regular monthly loaner exchanges. In the meantime, the loaner program is conducted from Kanch's home on a per-need basis.

SIG Imaging New Host

Hy Murveit introduced himself and said that he will be the new SIG imaging host taking over from Bruce

Electronically Assisted Astronomy's proposal for star parties

Swami makes a motion to approve reimbursement to Glenn for the purchase of equipment to be used for the Imaging program. In addition, Swami makes another motion to provisionally approve a budget for the purchase of 5 kits for the purpose of conducting public observing events.

July 24, 2021

In attendance

Swami, Vini, Glenn, Sukhada, Peter, Rob, Ken, Emi
Absences: Sandy, Wolf
Guests: Joseph, Leslie, Kanch, Sheena, Jai

Personnel changes

Ken announced that Emi has resigned from the position of Secretary. He, and others, thanked Emi for her efforts over the last 16 months as Secretary of SJAA. Emi will continue her duties as Ephemeris Editor.

SJAA Volunteer Coordinator

The Board reviewed and approved the set of proposed responsibilities and tasks for the SJAA Volunteer Coordinator.

Alternate/Backup Home for SJAA

The current negotiations with the city regarding use of Houge Park have been areas of concern. The board approved a proposal to create a task force to look into finding an alternate site that's not part of the City of San Jose, and then report back with a recommendation during the September meeting.

Monthly Newsletter

Jai volunteered to take on this task of creating a short, monthly newsletter issued shortly after the Board meeting that will highlight some of the "actions of interest" at the Board meeting, and other appropriate SJAA news. She takes over the task from Beth.

August 21, 2021

In attendance

Vini, Wolf, Sukhada, Peter, Rob, Ken,
Absences: Sandy, Swami, Glenn
Guests: Joseph, Leslie, Kanch, Sheena, Jai

Ongoing relationship between SJAA and the City

The City of San Jose is asking for details about our activities at the park. Rob stated that he'd like to see the proposed contract before making further outreach to SJ City. He stated that none of the community occupants at city parks have received or signed a contract.

Restarting SJAA public observing events

Wolf agreed to "get the ball rolling" by sending an email to the program leaders requesting that they each provide a description of their proposal to move ahead on this.

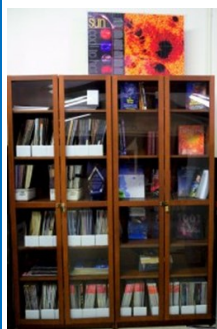
Alternate/Backup Home for SJAA

Sukhada volunteered to send an email to the membership asking for recommendations for potential alternate sites.

Reaching out

The board discussed making contacts to broaden SJAA's visibility by reaching out to schools, neighbors, the next door website, etc.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick START program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night

sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Program Coordinator for

additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir.	Ken Miura
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Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Peter Melhus
Director:	Vini Carter
Director:	Sandy Mohan
Director:	Wolf Witt
Director:	Swami Nigam
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Imaging SIG:	Hy Murveit
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In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
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Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Media Manager:	Beth Johnson
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SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

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Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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